Masroofy - Design to Code Mapping

This document explains the mapping between the Software Design Specification (SDS) and the implementation of the Masroofy Budget Management System.

Design to Code Mapping (Console Version)

SDS Class / Component	Responsibility	Code Implementation
User	Represents system user	class User
Category	Represents expense category	class Category
Expense	Stores expense details	class Expense
BudgetCycle	Handles budget calculations	class BudgetCycle
NotificationService	Sends budget warnings	class NotificationService
HistoryManager	Displays expense history	class HistoryManager
Dashboard	Displays financial summary	class Dashboard
Database	Saves and loads data	class Database
MasroofyApp	Main controller and workflow	public class MasroofyApp

User Story to Code Mapping

User Story	Code Implementation
Add Expense	expenses.add(new Expense(...))
View History	history.show(expenses);
Edit Category	expenses.get(id).category.name = ...
Set Budget	cycle.limit = sc.nextDouble();
Calculate Daily Limit	cycle.getDailyLimit();
Budget Notification	notify.check(getTotal(), cycle.limit);
Dashboard Display	dash.display(getTotal(), cycle.limit);
Save Data	Database.save(expenses, cycle.limit);
Load Data	loadData();

Sequence Diagram to Method Mapping

Sequence Operation	Related Method
User login	new User(sc.nextLine())
Add expense	new Expense(...)
Check budget warning	notify.check(...)
Calculate total spending	getTotal()
Display dashboard	dash.display(...)
Show history	history.show(...)
Edit category	expenses.get(id).category.name = ...
Save data	Database.save(...)
Load stored data	loadData()

UI Feature Mapping

UI Component	Related Logic
Dashboard Page	renderDashboard()
Add Expense Form	addExpense()
Expense History Page	renderHistory()
Budget Settings	saveBudget()
Budget Progress Bar	Updated in renderDashboard()
Warning Banner	Shown when budget >= 80%
Local Storage Persistence	localStorage.setItem()
Load Stored Data	JSON.parse(localStorage.getItem(...))

Architecture Mapping

Architecture Layer	Console Version	UI Version
Presentation Layer	Console menu using Scanner	HTML/CSS interface
Business Logic Layer	Java classes and methods	JavaScript functions
Data Layer	Text file storage	Browser localStorage

Notes:
- The implementation follows the SDS class and sequence diagrams.
- Both console and UI versions implement the same core functionalities.
- Persistent storage is supported in both implementations.
- The project follows Object-Oriented Programming principles and modular design.